

Comparing school leaders' experiences of research use in England, Scotland and Germany

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Abstract

In some countries it is common for policy makers to advocate research use by practitioners as a means to bring about school development. Yet, despite their increasing sophistication, the enactment of protocols for using research is problematic and, even in optimal environments, practitioners have difficulty mobilising research findings to improve student attainment. In this study, we consider school leaders' engagement with research, brokered by various intermediaries, in three countries where policy and practice differ. We explore research engagement through documentary analyses and in-depth interviews with school leaders from equivalent phase schools. Results suggest school leaders' experiences of research use are shaped by political contexts and strongly framed by education governance processes. These conclusions are, however, tentative and invite further investigation.

KEYWORDS

comparison, education policy, evidence informed practice, research use, school leaders

INTRODUCTION: RESEARCH USE IN SCHOOLS

The use of evidence by policy makers and practitioners to improve the quality of teaching is advocated by international organisations like the Organisation for Economic Cooperation and Development (OECD, 2019a) and often adopted in countries where education is a political priority and school success (based on student outcomes) is regarded as integral to their economic and social futures. Increasingly, the quality of teaching is a topic of discussion between

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Context and implications

Rationale for this study: This study seeks to understand the contextual factors that make research use difficult for school leaders by comparing documentation and school leader experiences in three countries.

Why the new findings matter: We identify how the contexts in which school leaders work affect their capacity to use research in school improvement.

Implications for policy makers, practitioners and researchers: This account is important for policy makers seeking to understand how policy interactions can affect research use. It will be of interest to school leaders and practitioners whose frustrations were evident in interviews, and to educational administrators, advisors and inspectors. All would benefit from understanding how professional circumstances can enhance or constrain research use in schools. Finally, an agenda for future research is provided.

policy makers and the wider public, creating a demand for evidence, be it standardised comparative data to hold schools and teachers to account, or research about what works to inform developments. But although national governments across Europe promote evidence-informed policy making in their efforts to improve schooling (European Commission, 2017), only a few engage school leaders and teachers in developing evidence-informed practice. Although the OECD (2019a) endeavours to make this more widespread, and despite some promise in the proposals of Maxwell and her colleagues (2022) in England and the Monash Q Project in Australia (Rickinson et al., 2023), 'it is easy to understand why universalistic recipes for school improvement of the sort promised by ... the OECD have proved elusive' (Furlong, 2022: i), when we consider how much teaching varies between countries.

In this paper we focus on one aspect of evidence-informed practice—research use by school leaders and teachers rather than data use—and consider how this is shaped by political context. We explore school leaders' engagement with research findings, whether in academic publications or brokered by various intermediaries, and their relationship with practitioner inquiry. However, current accounts of research engagement in schools ignore or underestimate how sociocultural and contextual factors make research use difficult for school leaders. To fill this gap, we consider how research engagement is promoted among school leaders and their experiences of research use. By comparing these experiences in three contrasting national policy contexts—England, Scotland and Germany—we explore how education governance affects practitioner engagement with research and identify the constraints on research use that school leaders in each policy context encounter. In so doing, we offer insights into the complexity of research use in different educational landscapes.

ENGLAND

The long-standing use of high-stake comparative student tests and inspections to evaluate schools and teachers in England is well documented (for example, Kelly et al., 2018). There is also an established tradition of using research to inform policy making and practice (Lawn & Furlong, 2010). Market reforms have opened up middle tier support for school leaders and teachers to commercial and philanthropic enterprises, including intermediaries who provide summaries of research findings to support decision making. These brokerage agencies

include the Evidence for Policy and Practice Information (EPPI) Centre, established in 1993, and Centre for the Use of Research and Evidence in Education (CUREE), which opened in 2008. Another, the Education Endowment Foundation (EEF), offers support to policy makers, and their *Teaching and Learning Toolkit*¹ has been available to schools since 2011. The EEF seeks to 'break the link between family income and educational achievement' (EEF, 2020) by (a) providing free, research evidence-based resources—summaries and practical tools—designed to help school leaders and teachers improve student attainment, particularly for those facing disadvantage, and (b) funding research into 'what works' in this regard. The Pupil Premium policy in England sits within this frame, offering a financial incentive to schools that admit children whose circumstances are often associated with lower achievement (DFE, 2010). Children from socio-economically disadvantaged backgrounds—in particular, those eligible for free school meals—bring additional funds, which schools must spend, specifically, on raising these students' academic attainment largely through research-informed initiatives. Ofsted, the school inspection agency, is tasked with assessing how well schools use research to inform their interventions. The National Institute of Teaching² is a more recent government initiative. This school-led organisation combines the role of broker and training provider (DFE, 2021). It offers initial and ongoing professional development programmes for teachers and school leaders that are informed by both existing research and their own studies of 'what works'.

SCOTLAND

Until the OECD review *Improving Schools in Scotland* (OECD, 2015a), Scotland's path had differed considerably from England's, with less emphasis placed on accountability measures and middle tier local administration maintained. Although policy highlighted the importance of research-informed practice, it sat alongside a strong tradition of teacher research in school improvement that valued partnerships between universities and schools. Following the review, however, discussions about school data as evidence increased and knowledge brokers were introduced to support practitioners' use of research in school improvement. Since then, policy documents such as the Research Strategy for Scottish Education (Scottish Government, 2017) explicitly recommend the collection, analysis and use of 'what works' evidence to improve academic standards among low achieving groups, and the EEF's *Scottish Attainment Challenge: Learning and Teaching Toolkit* was launched in 2017. Both position teachers as research-users rather than research-creators. The Toolkit accompanied the allocation of Pupil Equity Funding to specific schools, based on the socioeconomic disadvantage of their students, and was intended to support decision making about measures aimed at closing the poverty-related attainment gap. Although practitioner enquiry is still promoted (through, for example, the General Teaching Council Scotland (GTCS) standards and the Education Scotland model of professional learning), its omission from key education reform policies such as the National Improvement Framework (Scottish Government, 2022), the Muir Report (Muir, 2022) and the Scottish Government's Plan for School Research 2023–2026 (Scottish Government, 2023a) is notable. The Scottish Government's recent announcement to create a Centre of Teaching Excellence (Scottish Government, 2023b) has the potential to further solidify this shift.

GERMANY

The past 20 years have seen a move towards output evaluations in Germany. First, national standards were developed for each subject. Comparative standardised student assessments

administered by regional state (*Länder*) governments followed in 2009. Together with *Länder*-based school inspections, these are low-stake, less intrusive than in England or Scotland and have only a limited influence. Comparative data and inspection reports are only available to school leaders and local school administrators and provide formative feedback and inform school development. In addition, market reforms have been limited, and consequently schools enjoy less autonomy than in England. Nevertheless, 'what works' research is increasingly valued, and represents the emergent position (Schriewer, 2017). Beside the *Clearing House Unterricht*³ of the TUM School of Education, no national brokerage agencies of the kind that have flourished in England exist at present; there are 'functional equivalents' offering guidance on research such as the Leibniz Institute for Research and Information in Education.⁴ These cater mostly for advisors and administrators who provide middle-tier support at a federal level and have less direct influence on school leaders and teachers.

The OECD has been a key player in the spread of research-informed policymaking, and its more recent advocacy of research-informed practice is no doubt significant. Outside national accountabilities, it built on its administration and analyses of the PISA comparative tests to gain influence on national policy (Nordin & Sundberg, 2014). In Germany, for example, the shock of the PISA 2000 results (OECD, 2001) gave the move towards output evaluations considerable impetus (Ertl, 2006; Waldow, 2009). It has widened its role in recent years to provide explanations for national differences in PISA scores and act as a policy advocate for neoliberal reform (Sellar & Lingard, 2014), while its country reports, such as that on Scotland, have also been influential.

DIFFICULTIES WITH RESEARCH USE

Even in England, where some of the most enthusiastic advocates of research-informed practice reside, school leaders and teachers make little use of research, and evaluations of its contribution towards improving practice are unconvincing. Despite a growing demand for access to research findings among practitioners, 'it has proven difficult to achieve [research] evidence informed practice at a system level and the debate about how to achieve this continues' (Nelson & Campbell, 2017: 127). Coldwell et al. (2017) confirm that the influence of research-informed approaches on school development is limited. Walker et al. (2019) agree that, although teachers generally have positive attitudes towards research and indicate that their schools encourage this, research plays a relatively small role in influencing decision making. Gorard summarises the current position thus:

Much of education policy and practice takes place without concern for the growing evidence base. Despite admonitions and encouragement to take account of research, teachers and leaders still report that evidence is a minor factor in their decisions.

(2020: 17)

With sentiments like these in mind, some have explored whether it is feasible for teachers to use research findings to create their own interventions. During one academic year, See et al. (2016) evaluated how effectively teachers in nine schools in one local authority used research findings to create classroom interventions aimed at improving teacher feedback. They found little difference in student outcomes in these schools compared with other schools.

Supporters of research use have responded with a number of suggestions. They recommend access to research findings be improved. Cooper and her colleagues' exploratory case-study (2017) indicated that teachers tend to acquire information and ideas from those

they work with. LaPointe-McEwan and her colleagues conducted a longitudinal study (2017) which found that, although teachers use various data to inform their own practice, research publications have little influence. A systematic literature review by Hemsley-Brown and Sharp (2003) shed further light on this, suggesting that teachers are deterred from using research by too much jargon or high-level statistics, and Allen et al. (2007), reflecting on experiences of translating research into practice within the UK National Health Service, suggest that research findings are often expressed in such a way that it is difficult for potential users to discern what is relevant to their circumstances. Further, Siddiqui and Wardle (2020) argue that there are too many intermediary bodies presenting summaries of what 'the evidence says', which, they claim, is confusing for potential users. Alongside issues of access, others argue that teachers often regard research as irrelevant, but are more likely to engage with studies focused on teaching, learning and improving practice (Cordingley, 2000; NTRP, 2006).

Scott and McNeish (2013) suggest that school leaders and teachers regard research as credible when they feel that the suggestions that arise from studies are realistic. They recommend practitioners are actively involved in exploratory and adaptive practice as they engage in constructing interventions. To do this they need support in translating research findings into useable practices, and most importantly, the time skills and confidence to do so, in schools that are supportive of their endeavours.

Following their quasi-experiment intervention-based study conducted across ten schools, See et al. (2016) call for professional learning focused on research, to increase school leaders and teachers' capacity to use research. In this regard, Carrier (2017) identifies the need for school leaders and teachers to develop analytical and evaluative stances towards using research findings and school data to improve practice. Within a conceptual analysis of previous literature on data-based decision making and evidence informed practice, Brown and his colleagues (2017) advocate for the development of research and data literacy and suggest that both are important if teachers are to engage in professional enquiry and school improvement. Following empirical research, conducted through quantitative surveys with teachers, Brown and Zhang (2017) task school leaders with developing teacher capacity, promoting school cultures that recognise the importance of research use in effective learning environments, and putting effective structures, systems and resources in place that facilitate research use and the sharing of best practice.

Some offer solutions that complicate school improvement processes. For example, Brown et al. (2017) argue that data-based decision making and research-informed practice are both crucial for educational improvement, but are under-developed because they are often employed independently of each other. They propose their complementary use, with the strengths of one compensating for the weaknesses of the other and set this within a systematic cycle of inquiry focused on improving student learning. Similarly, Schildkamp (2019) provides a sophisticated model for how school leaders and teachers might use a variety of data in decision making, including summative assessment data, formative assessment data and educational research findings, to improve the quality of education. In another example, following a two-year project in England, Greany and Maxwell (2017) suggest that collaborative research and development can facilitate research informed innovation and provide a framework to support this. The EEF (2019), whose online research evidence platform was evaluated as having only a moderate to small influence on schools (Nelson et al., 2017), has offered an approach to research-informed school improvement in the form of a five-step cycle. In this, they regard schools as continuously improving systems. The Research Schools Network (Dixon et al., 2020) is another initiative in England that aims to develop strategies and tools to help schools engage effectively with research. It claims to have contributed a number of studies that illustrate how to support schools effectively, but none have been properly evaluated.

In summary, the logic—and, indeed, the hope—underpinning all of these suggestions is consistent; clearly reported relevant research findings can be used by skilled practitioners working in supportive school communities to inform their practice and enhance student learning.

ACKNOWLEDGING COMPLEXITY

Systems perspectives such as the EEF's five-step cycle may be more helpful for understanding and improving research use in schools because they account for some of the complexities ignored by others, including the social and political conditions in which schools operate and practitioners work (Rickinson et al., 2023). They recognise the multiple actors involved, from research producers to brokers and school administrators to practitioners, suggesting that the efforts of all need to be aligned for research use to be successful (Maxwell et al., 2022). And they invoke the notion of complexity to understand how the consequences of small changes can appear disproportionate (Cairney & Geyer, 2017). There is a functionalist perspective, where each part of the system acts like a cog in a mechanism, their actions combining to bring about effective research use. The behaviours of school leaders and teachers are driven by the demands of their roles in schools, while schools strive to serve wider social purposes. Such intentions are aspirational and relate to the private, public and common goods of education (UNESCO, 2018). Hence, preferred outcomes shape the means used to realise them in functionalist analyses; a reversal of more conventional analyses that explore the intended or unexpected consequences of actions.

One example is the Monash Q project (Rickinson et al., 2023). This identifies different enabling subsystems, including 'spaces that productively nurture research use' and 'approaches to supporting and monitoring this', and looks at how each can be optimised. As in earlier examples, the primary means to improve research use involves developing the skills of individuals, their dispositions towards research use and the organisations in which they work to enable this. Elsewhere, Maxwell et al. (2022) identify the role that intermediaries at different levels can play in facilitating research use. Both accounts require the rational actions of key players for their enactment.

As is common to many systems theorists, these projects employ the metaphor of self-regulating and self-righting ecosystems oriented towards clearly identified aims, a position that highlights their conservative nature and proclivity for gradual and adaptive change. The capacity for continuous improvement lies within such systems and is enhanced when the actions of individuals are aligned and optimised to increase their effectiveness and efficiency in meeting desired outcomes. However, this focus on individual capacities to affect change ignores the insights of sociocultural accounts of practice, which we discuss shortly, that identify barriers to knowledge mobility. With little attention given to the impact of power imbalances, the patterned but unintended privileging and marginalisation of particular groups and perspectives is also ignored. Furthermore, they provide few opportunities to recommend broader changes or more extensive policy overhauls.

THE ROLE OF SCHOOL LEADERS

The prescriptions provided in this literature rely on the activities of school leaders who, authors assert, play a central role in the most highly research-engaged schools (Coldwell et al., 2017), show the highest levels of research engagement and knowledge (Nelson et al., 2017), and are therefore regarded as important in expediting research use

(Brown & Zhang, 2017). This is not just school principals. Skerrett and his colleagues (2021) suggest that, whereas head teachers set the overall direction of travel for schools, it is often middle leaders who bring policy to life by organising, managing, leading, planning, producing, inspiring, persuading and appeasing colleagues. They reported how middle leaders in Ireland play a central role in encouraging professional collaboration and translating policy into practice (Ball et al., 2011, 2012) even as they are inundated and often overloaded with other demands.

Such debates about how best to promote research use sit within an instrumental-rational frame (Wilkins & Gobby, 2021) that lends itself to functionalist accounts of educational leadership as the strategic management of organisations seeking to improve their performance on output measures, often within contexts that vary in their degree of marketisation and accountability. However, some proffer alternative reasons for the disappointing uptake of research use by practitioners. A number identify limitations in the selected research to suggest that the medical model underpinning research informed practice is inappropriate for education (Biesta, 2007; Hammersley, 1997). Others add that randomised control trials and systematic reviews, the primary methods used, are inadequate and insufficient (Wrigley, 2018). Finally, there is recognition that using research-generated knowledge in practice is extremely difficult (Biesta, 2007, 2010; Cain, 2015; Hammersley, 1997). As such, Philpott (2017) warns against uncritically assuming the superiority of medical models of professional learning or relying only on empirical evidence to inform practice. Indeed, Hemsley-Brown and Oplatka (2005) suggest that the barriers to research use in education are greater than in other contexts because of the 'uncertain, ambiguous nature of teaching and schooling that makes it difficult for researchers to identify clear, valid principles and findings based on hard evidence' (2005:24). They conclude that research utilisation in social professions is likely to be lower than in healthcare.

For others, the underpinning assumption of considered and deliberate choice, which lies at the heart of functional analyses of school leadership, is mistaken. Bourdieu critiques rational choice theories (Wacquant, 1989) that emphasise individual and reflective decision making because they overlook collective dynamics and histories, the affordances and constraints of circumstance and the unconsidered, impulsive and sometimes irrational activities that together constitute social life. In this vein, Allen et al. (2007) argue that, for many potential research users, the day-to-day pressures of running an organisation and responding to multiple demands and targets are likely to override any desire to make decisions using research. Meanwhile, Wilkins and Gobby (2021) provide an agnostic-political view that regards educational leadership as a site of struggle. This is evident when school leaders adjudicate between competing priorities, some representing incommensurate values, as they: negotiate how to forward the interests of a diversity of students; take into account the contrasting views of parents and other education professionals; while also meeting mandated requirements.

ADVOCACY AND PRACTICE

Following similar lines to these later critiques, a recent analysis of expertise in the public sphere (Eyal, 2019) identifies two tensions between expert advocacy and expert practice. The first contrasts the technical objectivity of academic researchers and intellectual commentators whose materials are used to inform policy making and practice with the trained judgement of participants and decision makers in the fields where policy and practice are enacted. Technical objectivity is characteristic of interactional experts (Collins & Evans, 2007), so called because of their ability to persuade others of the veracity of their claims. In their advisory roles, they largely assume cognitive models of expertise;

tending to focus on individual actors as their units of analysis, and following reductionist logics to identify abstractions that, they assert, can be subsequently applied to specific contexts and circumstances. This differs from the trained judgement of participants and policy makers on the ground. These contributory experts (Collins & Evans, 2007) readily overcome challenges within specific contingencies to get things done. Their activities are better captured by sociocultural accounts that acknowledge the embedded and embodied nature of practical competence; one which draws heavily on tacit and local understandings learnt through many years of experience and particular to the contexts and circumstances in which they were acquired. This hints at a second distinction; between technocratic and democratic decision making. Technocracies rely on singular authoritative accounts of the social world provided by individuals with the capacity to understand and apply these to everyday concerns. Democrats, however, understand that such knowledge is contested, fallible and transient, and favour collaborative and deliberative decision making that draws on diverse perspectives and experiences.

By implication, whereas the legitimacy of contributory experts rests in their practical success, the credibility of interactional experts hinges on how convincingly they express, defend and promote their claims. Nevertheless, experts from both standpoints secure trust by asserting their disinterested promotion of the welfare of others, even as the actors involved use their resources, status and alignment with socially valorised positions to increase their persuasiveness.

RESPONDING TO THE EXISTING LITERATURE

Despite much support, education professionals continue to find it difficult to use research findings to inform their decision making. Increasingly, approaches have acknowledged the complexity of research use and become more sophisticated. However, the models underpinning these approaches remain some distance from the everyday circumstances in which school leaders and teachers often work. For example, they ignore the many often incommensurate demands placed on headteachers and pressures from high-stakes accountabilities. Nor do they question whether workplaces allow employees the time and space to make rational choices. Yet, even in more conducive environments, school leaders and teachers may have difficulty incorporating research findings into professional knowledge that is socially embedded and individually embodied. In this study, we set out to explore this further by looking at research use from the perspective of school leaders in three contexts where their working circumstances differ. To do so, the study sought to answer the following research questions:

1. How is research engagement promoted among school leaders in three contrasting European policy contexts?
2. What constraints on research use do school leaders in each policy context encounter?

METHOD

Engagement with research can involve school leaders using findings, accessed directly from academic publications or in summaries brokered by intermediaries, to inform their decision making, but can also include them or their colleagues undertaking small-scale studies with the aim of improving their practice. To understand these better, we analysed relevant documentation and the experiences of school leaders in countries selected for their policy diversity.

Documentary analysis

As one intention of the documentary analysis was to explore the governance of research use, we accessed government documents from two bodies in England, the DFE and Ofsted. This included the school inspection framework (Ofsted, 2019a) and a report of the research underpinning it (Ofsted, 2019b). Given structural differences in Scotland, however, we only looked at documentation from the Scottish Government, which included plans for improvement (Scottish Government, 2023a) and ideas about educational development (Muir, 2022). Educational governance is regional in Germany, and government documents from the city-state of Hamburg were considered, as this is where the school leaders who participated in this study were based. Among the documents analysed was the state's school quality framework (Schulinformationszentrum, 2019).

Next, we searched the websites of international non-governmental organisations, specifically the European Commission, OECD and UNESCO, and of organisations that encouraged and supported research use by school leaders in each country, including important research brokerage agencies such as the EEF in England and Scotland and the Leibniz Institute for Research and Information in Germany. On each website we identified web-pages with policy briefings, reports and guidance for school leaders and other educational professionals.

We completed the documentary search by following a snowball approach. As the interviews proceeded, documents and web materials from intermediary organisations that were mentioned by school leaders but not amongst those we had already identified were added. Similarly, when new documents were identified, references in them to new materials were followed up. Fortunately, the total number of documents for each country and in each category remained manageable. The number of documents included from non-governmental and governmental sources is shown in Table 1. The numbers of websites analysed for non-profit and commercial intermediaries is also shown. The totals reflect the interest in research use by school leaders in each country. Finally, the identified documents and web materials were categorised according to their target audience, research selection and presentation, and the approaches offered to support utilisation. When not referenced directly, the web materials analysed are included as footnotes in the sections that follow.

Exploring school leaders' research use

Next, we conducted hour-long semi-structured interviews with three school leaders in each country. Participants were recruited from three regions identified as geographically and socially similar: the Hamburg Metropolitan Region, the Glasgow City Region and Southwest England. Their characteristics are identified in Table 2.

TABLE 1 Sources of the documents that were selected for analysis.

	England	Germany	Scotland
International non-governmental organisation	0	2	3
National or regional government	4	4	5
National non-profit intermediary	7	5	4
Commercial organisation	8	2	5
Total	19	13	17

TABLE 2 Information about the participants.

Participant	Location	Role	School	Time in role
A	England	Headteacher	Large, non-selective secondary school for students aged 11–18 years, situated in an urban area of mixed deprivation	10 years
B		Headteacher	Special school for students with severe and complex needs aged 3–19 years, situated in an urban area	Over 10 years
C		Headteacher	Large primary school for students aged 4–11 years, situated in an urban area of high deprivation	5 years
D	Germany	Headteacher	Large, non-selective secondary school for students aged 10–18 years, situated in an urban area of mixed deprivation	Less than 5 years
E		Senior Leader	Large, non-selective secondary school for students aged 10–18 years, situated in an urban area of mixed deprivation	5–10 years
F		Headteacher	Large primary school for students aged 6–10 years, situated in an urban area of mixed deprivation	Less than 5 years
G	Scotland	Headteacher	Medium-sized primary school for students aged 4–11 years, situated in an urban area of high deprivation	Over 10 years
H		Headteacher	Medium-sized primary school for students aged 4–11 years, situated in a rural area of mixed deprivation	5–10 years
I		Headteacher	Special school for students with severe and complex needs aged 3–18 years, situated in an urban area	Less than 5 years

In these interviews, we explored school leaders' understanding and use of research to inform their practice, focusing on (a) the research and sources of research that school leaders could identify as being available to them; (b) their assumptions, understandings and experiences of research engagement; (c) their accounts of how research engagement has affected practice; and (d) their views on research usefulness. The questions we used are included in [Appendix 1](#).

Ethical procedures relevant to each country were followed at all times. Participants were provided with information about the nature and purpose of the study and their rights to anonymity and withdrawal and signed a consent form to agree to take part. To maintain appropriate cultural awareness and sensitivity, insider researchers, fluent in the national language, and familiar with both the broader policy context and local circumstances, conducted documentary searches and interviews and undertook the initial analysis of each. School leaders' responses to the three areas identified above were grouped according to emergent categories, whilst the documents identified beforehand and by the school leaders themselves were used to contextualise their answers. The research team then brought their initial categorisations and understandings of these as both insiders and outsiders to the different countries being examined, together to identify similarities and differences in a process of comparative analysis.

As the documentary analysis serves to contextualise the reported experiences of school leaders in relation to research engagement and constraints on research use, these are combined in the account that follows.

RESULTS

School leaders' research encounters

Although only John Hattie's *Visible Learning*⁵ was mentioned by school leaders in all three countries, international organisations are increasingly influential in Germany and Scotland. For example, variations on PISA such as *PISA4U*⁶ are gaining significance, as is the OECD's role more broadly in brokering or interpreting its own data as evidence to inform its own guidance on improving policy and practice. However, in England, the numerous national actors promoting research have largely crowded out the work of such organisations.

Promoting research engagement in England

The Department for Education and government agencies like Ofsted, commercial, semi-commercial and philanthropic middle-tier organisations and other intermediaries, along with some local authority provision and services provided by multi-academy trusts all act as arbitrators between researchers and school leaders. These brokers include the aforementioned CUREE, EEF and EPPI-Centre, the Research Schools Network⁷ and subscription communities linked to professional training or networks that support research-informed school improvement, like the Chartered College⁸ the Institute for Teaching⁹ and ResearchED.¹⁰ But despite the pressure placed on them, school leaders say they rarely have time to engage with research, and on those occasions when they do, they admit it is with a degree of naivety. Instead, they rely on the knowledge and judgement of others claiming expertise in this area; most often, advisors, consultants and inspectors. School leaders also report subscribing to or purchasing outright curriculum schemes and resources from businesses and philanthropic organisations servicing schools, one being *Thrive*,¹¹ which publishers say are based on research, thus affording their materials credibility and rendering them trustworthy.

As such, state, commercial and third sector intermediaries are proactive and highly influential in selecting, presenting and interpreting research for use by school leaders.

We are inundated—there's something new every single week—an idea, a company promoting an approach, a government guidance document, something from a subject organisation or internal things from different people. You are constantly in a situation where you have to decide what to look at and what to ignore and when things look like what's gone before, you tend to ignore it.

(School Leader B)

For one school leader, any research discussed at staff meetings or wider networking events, 'mostly comes from the EEF, and that is what head teachers drop into what they say at professional meetings to add authenticity to what they are doing at school. This is also the case with other agencies we work with, who justify what they say using the EEF' (School Leader A). Meanwhile, many school leaders and teachers also bring in ideas from the education news media; as one suggests, 'teachers also come across mainstream research from the TES [the *Times Educational Supplement*, a weekly educational newspaper] and then they may go and have a look at it—occasionally back to the original papers, but more often to summaries of their work on websites' (School Leader A). However, awareness of research is not universal: 'only half of heads I know of use research to inform their [Pupil Premium] spending or even know about the EEF' (School Leader C).

Clearly, school leaders in England have an urgent need to use research to inform practice; both to improve student outcomes and to benefit from the professional credibility from inspectors and parents that research use bestows. This creates a division between those with the capability, inclination, time and wherewithal to do so to their advantage, and those lacking in some or all of these. The resulting market opportunities within the disaggregated English context have attracted numerous knowledge entrepreneurs; advisors, brokers, consultants and their interpreters, working for state, philanthropic or commercial organisations. Together with researchers, they comprise a knowledge elite (Ozga, 2020) of interactional experts (Collins & Evans, 2007) who present evidence in the form of generalised principles, accounts of what works or examples of best practice, and design services and resources to make these directly actionable by policy makers, school leaders and teachers striving to improve practice.

In all cases, medical research acts as the model for research informing practice as a 'new science of education' (Furlong & Whitty, 2017). Typical of this is the overview of research (Ofsted, 2019b), mentioned by all interviewees, which underpins and was published alongside the most recent Ofsted *Education Inspection Framework* (2019a). The research used in this review aims to identify 'what works' in schools to raise student attainment through the application of methods aligned with evidence-based and evidence-informed-medicine, and mirrors the research promoted by other intermediaries. Typical studies take the form of randomised controlled trials and systematic reviews, but also include national and international school effectiveness research and disciplinary inquiries in areas such as neuroscience and psychology, which employ statistical and scientific methods. The agenda for this research is largely set by funders, and the institutions carrying out commissioned research are increasingly commercial rather than university based (Gorard, 2020). Consequently, clinical models of professional development are favoured (Furlong & Whitty, 2017; Philpott, 2017), such as that advocated by the EEF in *Putting Evidence to Work: A School's Guide to Implementation* (2019), which focuses on supporting school leaders and teachers in the collaborative use of such research to improve practice. In support are what Furlong and Whitty (2017) call learning sciences approaches. Although interdisciplinary, these favour studies that use scientific methods—largely in statistical, neuroscience and psychological research—and exploit their findings to design and improve learning environments. Hattie's *Visible Learning*, mentioned earlier, synthesises the findings of quantitative research studies within a Piagetian frame, and Guy Claxton's *Building Learning Power*¹² is another, based largely on neuroscience. Meanwhile, functionalist and techno-rational management approaches also dominate. Organisations such as the National College of School Leadership, which became the National College for Teaching and Leadership and closed in 2018, set the tone by pushing largely instrumental management research literature, which the Department for Education continues to advocate.

Promoting research engagement in Germany

Head teachers in Germany consider the turning of external research and expertise into professional guidance to be part of the role of federal administrators. Instead, they focus on using the information they receive from educational monitoring, testing and inspection, provided by local administrations such as the Institute for Educational Monitoring and Quality Development, which belongs to the Hamburg Department for Schools and Vocational Education, to identify areas for improvement. Brokers such as the Leibniz Institute for Research and Information in Education in Frankfurt and the *Clearing House Unterricht* in Munich provide research knowledge on a national level, but they are rarely noticed beyond

the federal state. Consequently, school leaders' direct engagement with research ideas and publications is fairly limited. Although reports of research studies are available online through organisations like the Bertelsmann Foundation,¹³ they are little used by school leaders. With the exception of Hattie's materials, few head teachers know about learning science resources from commercial and international organisations, and even fewer use them. When asked directly, most school leaders cannot name either publications or platforms such as the *Forschungsmonitor Schule*,¹⁴ although one has used self-study modules and networking opportunities provided by the Digital Learning Lab.¹⁵ This 'open online competence centre for teaching in digital times' is run by a consortium of *Länder* administrators and technological organisations including the Hamburg Technical University. Another has consulted a handbook on professionalism and teaching quality at schools, written by an educational researcher (Helmke, 2017).

Professional training addresses mainly functionalist management ideas and, increasingly, diversity and intercultural education since the migration crisis of 2015. For this, considerable support is provided by the federal states, and school leaders reported that training and conferences had a lasting and helpful effect.

In general, I use sources of information provided to me by the state institute and from training events that I attend. New or newly appointed head teachers attend a series of training events that deal with different aspects of school development, from personnel management to legal aspects and so on, and you get a reader that summarises everything and indicates the sources of ideas. Later, you can read and follow up with a few basic texts.

(School Leader D)

Although one consulted an advisor with knowledge about student behaviour problems, more often school leaders talk of the benefits of informal discussions with colleagues, when suggestions for change and development are identified. Teachers bring ideas to these from professional networks, publications and training or the education news media.

We considered support from the state institute and from outside the classroom. So, we asked, what do we do when things go wrong, who is responsible for what, who is the right contact person—guidance counsellors, special needs teachers—and which external agencies can and should be consulted?

(School Leader E)

Sometimes local administrators provide the impetus for cross-school collaborations:

Our school board sets a theme each school year, such as 'the development of teaching' or 'inclusion', forms a team and selects possible sources for ideas and actions in a goal-oriented manner.

(School Leader E)

At other times, in-school concerns are followed up:

The impetus for considering inclusion issues came from a colleague concerned that the families of children from disadvantaged households were hard to reach. A regular questionnaire and conference-based assessment [called the *Gefährdungsanalyse*¹⁶] identified such challenges as a source of workplace stress.

(School Leader F)

Informal discussions about new publications, texts and research results are also emphasised:

And quite a lot happens on an informal level through colleagues who bring in new input and say, 'here's something exciting' or 'here's a text'. And that moves both the management team, we are five, so actually six people, and [non-management] colleagues.

(School Leader F)

In this secondary school, teams of 'teaching improvement groups' have been formed that read, discuss and summarise selected research from Hattie's publications to identify helpful ideas for their colleagues. Media reports also act as a stimulation for some, although others reject such conversations.

We also have colleagues who say, 'leave me alone'. [Nevertheless], there are colleagues who will certainly read an article. I would say that there are few colleagues who really look at the studies ... and read them through completely.

(School Leader F)

Characteristic of all of these examples is the way school leaders and practitioners combine their contributory expertise (Collins & Evans, 2007), situated craft knowledge (Furlong & Whitty, 2017) acquired through years of experience and developed through reflective practice and collaborative learning activities, with ideas from a diversity of interactional experts.

Promoting research engagement in Scotland

In Scotland, research findings are part of a coherent and centralised support framework offered by government and their agencies. Together with local authorities and the GTCS, they promote a range of research models. This includes new science of education medical-like research similar to England, which shares information about 'what works', programmes based on interdisciplinary learning sciences approaches, practitioner enquiry models (Furlong & Whitty, 2017) with explicit links to practice, and some critical research based in disciplinary understandings and accessed through postgraduate study and professional learning events. As in England, these are brought together and employed in a clinical development model, although the base this draws on is much wider, and throughout, the applicability of research to classroom practice is closely linked to perceptions of its usefulness.

The government has provided Pupil Equity Funding (PEF) directly to schools since 2017 as part of the *Scottish Attainment Challenge*, marking the first time that government funding has bypassed local authorities. However, although school leaders have witnessed an increase in the number of direct approaches from businesses and philanthropic organisations, and are able to purchase programmes and support directly, local authorities have retained a powerful role. Until PEF, it was common for research intermediaries and brokers to go through local authorities to gain access to schools, rather than marketing to schools directly, and research-informed programmes from non-profit and third sector organisations continue to be purchased by local authorities and distributed to specific schools. One school leader identifies how a behaviour support programme, offered by PATHS¹⁷ was presented to them as a 'huv-tae' by their local authority: "if you had a high exclusion rate, you had to do it" (School Leader G). Local authorities also organise professional

learning events targeted towards school leaders with well-known academics as invited speakers including Andy Hargreaves, John Hattie, Dylan Williams and Michael Fullan. Hargreaves also has a position on the Scottish Government's International Council of Education Advisers¹⁸ and close connections to the University of Glasgow.¹⁹ Similarly, learning science programmes developed by well-known academics including these are purchased by local authorities and either offered directly by academics and consultants, or through organisations like the Tapestry Partnership.²⁰ But school leaders sometimes act independently; one purchased and implemented a Therapeutic Crisis Intervention²¹ developed by Cornell University.

School leaders and intermediaries agree on the importance of research accessibility, and most platforms provide a combination of simplified or summarised versions of original research findings and examples of their practical application. The GTCs provides access to research for all registered teachers through the online database Education Source, from EBSCO Information Services.²² However, one school leader warns, “you maybe get a lot of abstracts, but not full articles” (School Leader I). Education Scotland, a governmental agency responsible for curriculum and inspecting schools, also provides summaries of research through its National Improvement Hub,²³ which aims to ‘improve outcomes for learners’. However, searching the hub will yield an array of documents, websites and resources, not all of which are evidence-based. School leaders cite this as one way to access research, but none of them report having used it. Education Scotland also provides access to *Interventions for Equity*,²⁴ which has summaries of practitioner enquiry projects, case studies and examples of ‘good practice’ from various local-authority-led programmes²⁵ supported with references to evidence from research and classroom practice. The EEF's *Toolkit* is offered as part of this but appears towards the end of a long list of resources. When it was first introduced, a Scottish-specific version with examples from practice to ‘reflect the specific needs of the Scottish education sector’ (EEF, 2017) was promised but failed to materialise. Instead, a link is provided to the standard version that is used in English schools. The toolkit's somewhat insignificant position reflects the flexibility of the national and local approach to research use in schools.

A key difference between Scotland and England is that, although various forms of research are promoted by local authority and government agencies, there is flexibility for schools to decide which type they want to use, depending on teacher's judgement about what fits their context best. As observed by one school leader, ‘we have really been given freedom to do what we think is right for our communities’ (School Leader G). Social media and the TES are also used by school leaders to keep up to date with recently published research, with the latter referred to as a ‘major gateway’. They also engage with Pedago, a teacher-led professional learning network, and ResearchEd.

School leaders and teachers as researchers

In England, a small, important but increasingly marginal group of school leaders that includes those undertaking postgraduate study, engage directly with social research ideas and practitioner research approaches in trying to improve professional practice. Additionally, ‘one or two heads read books and popular educational materials in their own time. When I started, we had time to read things and try different things out, but these days it seems to all come from above’ (School Leader B). Similarly, while school leaders in Germany indicate that they have no time for and little interest in conducting research themselves, and most only learn about research informally, a few have colleagues occupying hybrid position as teachers and university-affiliated mentors for student teachers, lecturers or doctoral candidates, who engage with research. But teachers with a personal interest

in research engage with it in their own time, as schools have no resources to support this, leading School Leader E to say, 'it is definitely a big burden for these colleagues. You have to say that very clearly, they do not have any advantage from it.'

However, school leaders from Scotland are much more positive about the benefits of university partnerships and are partly influenced by a critical disciplinary literature although universities also provide 'evidence-informed' training programmes. Indeed, universities are listed as the easiest and most effective way of accessing research by many school leaders who currently or recently have engaged in postgraduate study themselves. Recalling a time when they did not have the same level of access, one sympathises with colleagues unable to access full versions of journal articles readily: 'first of all, you don't know where to look, and secondly, it has cost implications'. But another school leader bypasses this by contacting researchers directly to request access to their work: 'if I can't find it, I will email them and they are very good at sending them' (School Leader I). Yet others, 'wouldn't dream of emailing an academic to ask for a paper. It's that hierarchical thing' (School Leader G).

Additionally, a number of school leaders engage with practitioner enquiry and promote this with teachers in their own schools. Spurred by its centrality in GTCS Professional Standards, this is increasingly popular for professional learning and school development (Chapman et al., 2016), and has become a core feature of the induction year, with probationary teachers required to conduct their own enquiry projects. These enquiries can have a wider impact as, for one school leader, they 'nudge' his learning: 'It led head teachers into a deeper understanding of what research is about'.

School leaders and teachers as research users

Accounts of research use in England

In England, an abundance of actors encourage busy practitioners to use research to improve student outcomes and meet Pupil Premium requirements. The government, its agencies and middle tier organisations together promote research use in schools, reminding school leaders about the importance of using research findings to guide approaches to school development in general and their response to school data in particular, whilst philanthropic intermediaries and research brokers select and translated research for school leaders and teachers to use directly in decision making. Meanwhile, numerous commercial providers use research to design instruction and behaviour management approaches for schools to subscribe to or purchase, often using Pupil Premium funding. Of these, Hattie and Claxton's work remains influential.

Building Learning Power was introduced through a countywide conference for head teachers. They were speaking about and promoting the materials at the conference. We decided to go with that as so many other schools were doing it and saying how good it was. In ways like this, groups of schools look at what each other are doing and what is effective and then it becomes popular.

(School Leader A)

Teachers working with students with special needs and disabilities have long used resources and tests but, in recent years commercial organisations have begun offering more comprehensive services. This includes *Thrive*, a popular research-based subscription approach for dealing with behavioural problems that also provides resources and professional

training and support. On a smaller scale, some schools have themselves begun offering behavioural support services that they have developed to others.

I know that two other schools ... are developing nurture group models for short-term placements for children aged 6 to 13,²⁷ who find mainstream education challenging at times. They say it is evidence based, but not I'm not sure what the evidence it is based on is.

(School Leader B)

This is also the case for large multi-academy trusts like Ark,²⁸ a network of 38 schools who offer subscriptions to ready-made detailed schemes of work, teaching programmes and associated resources in subjects like English, science and mathematics,²⁹ which they claim are research-based, and sitting alongside curriculum service providers are subscription networks such as Challenge Partners,³⁰ who have sought to capitalise on opportunities for inter-school research and development collaborations rooted in clinical models of professional development. Another example that one school leader participates in is the *Challenge the Gap*³¹ project, marketed as a strategic, evidence-based model by which schools develop new practice and share with each other what works in order to break the link between poverty and underachievement.

Accounts of research use in Germany

In Germany school leaders tend to conflate all of the information available to schools. This includes, on the one hand, statistical, comparative and evaluative data and reports, and on the other, professional guidance and commercial learning science materials claiming to be based on research. They identify both formal and informal support for school development but are unconcerned about whether this counts as research. As such, the research label carries little weight and does not compel school leaders to attend to it more than any other of the materials available. Instead, they trust their own professional judgement when deciding on the usefulness of any information provided and how to make use of it.

It is important to distinguish between textbooks, which are used extensively to support teaching and are based on an analysis of the subject and arranged to ensure curriculum coverage, and learning science materials, which advise on the form of teaching. Hattie, one of very few mentioned, does not so much inform practice as act as a starting point for teachers' own professional explorations.

Hattie was an issue, indeed, because of the question of class composition, class size, and the importance of homework, we picked up many developments. Moreover, these were taken up by a working group, and the working group then presented the results at an overall conference. That was important. In addition, we have ... in terms of teaching development on the quality of teaching ... a tool used from the literature. And that's the point, so the idea was for colleagues to visit each other to see best-practice examples, talk about lessons, and drive lesson development.

(School Leader D)

But more often, it is professional education and training and head teacher conferences that are the main source of information and research knowledge, as well as local guidance such as the Orientation framework for school quality (Schulinformationszentrum, 2019) in Hamburg.

Accounts of research use in Scotland

All three Scottish school leaders identify the learning science programmes and materials they use as examples of research informing practice. Each is associated with well-known academics Claxton and Hattie, mentioned already, as well as Ron Ritchhart, Mark Church and Karin Morrison's *Making Thinking Visible*,³² Andy Hargreaves and Michael Fullan's work on professional capital and Dylan Wiliam's work on formative assessment. Well-known academics including these are often invited to speak at national and local conferences and regarded by school leaders as 'knowledgeable experts'. But some voice concerns about how their ideas are translated into practice:

What worries me about this is what happens is you get key messages taken out of context.

(School Leader H)

School leaders are also aware of a tendency for the uncritical consumption and overuse of such materials because they had been 'cleverly marketed':

Things come in big cycles, and fashions and trends, and it is the next big thing. The Hattie stuff is a good example of that—*Visible Learning*—all of a sudden everybody's talking about it.

(School Leader G)

To avoid this, one school leader uses practitioner enquiry to gather information on the impact of changes based on *Visible Learning*, and to secure buy-in from teachers, suggesting this is 'a good way to get teachers on board' (School Leader G). Indeed, school leaders stress the importance of getting feedback from teachers on the implementation of programmes.

Some like to be seen as interested in research and are aware that they: 'need to be careful about creating that space to allow them [teachers] to do that [engage with research]. It has to come from the top' (School Leader G). Yet, School Leader I admits, at times, it feels like an 'add on'. Nevertheless, school leaders encourage teachers to engage with many sources of research literature including articles in professional newspapers such as the *TES*: 'we start every staff meeting with a professional reading, I try and bring something that is controversial' (School Leader H). Another suggested research should be introduced gradually to avoid it appearing 'threatening', adding, 'spoon feeding, here is a bit of reading on this, here is a video on that, works for my staff' (School Leader G). One school leader encourages collaborative working to implement research ideas:

It was a paper by Fullan on collaboration. I gave that out to them all first and we read through that, and I asked the other heads to make sure that all the other teachers got that too.

(School Leader H)

Some admit to making instrumental use of research. School Leader H uses educational research as a 'platform' or as 'a bit of weight—so it's not just the world according to me'. Here, research is mobilised to justify arguments and encourage teachers to get on board with new actions. However, there are constraints to the extent to which school leaders and their colleagues can engage with research. Time is a constraint for all three school leaders, with one asking, 'where do you actually get time? Everyone is so busy spinning plates. Sometimes you are just trying to get through the day' (School Leader G), while for School Leader I, 'you need that head space that allows you to do that'.

School leaders' thoughts about research usefulness

Benefits of and constraints on research use in England

The English school leaders considered research useful if initiatives informed by it could be linked to improved student outcomes but cited a lack of time as the biggest obstacle preventing greater engagement with research, as well as an inability to engage meaningfully with the studies they encounter:

Most heads don't really know what the criticism of this kind of research are, and so they just take it at face value—but it is the same with teachers too. People really don't have the time to access materials, and so they just pick up snippets, soundbites or summaries provided by other people. Or they buy into things that are supposed to be based in research.

(School Leader A)

With so many competitors offering suggestions, services and resources, school leaders report that they are overwhelmed and, given their workload, this means decisions on which research to use or services and resources to buy are often made following personal recommendation and discussions in professional networks. Practitioners try out new informational sources or programmes recommended by advisors, consultants or colleagues, both in formal professional development and during informal networking activities but are sometimes suspicious of well-known evangelists:

Research is always pushed by people promoting their own materials or trying to offer you professional development.

(School Leader A)

Practitioners also use the EEF and other toolkits, which supplant processes of reflection:

We don't really talk about reflective practice anymore—it is more, look at the data, find out where we can improve, look at what someone says research tells us about how to do that and then buy in the resources or make the changes.

(School Leader A)

Research findings offered by these intermediaries are less trusted when they contradicted personal experience:

We looked at the EEF stuff on teaching assistants and whether they were value for money, but with Thrive assistants are trained and use it with one or two students and they do benefit and so I am not sure that the research is very good for specific things like that.

(School Leader B)

For some of the more research engaged school leaders, however, research use is instrumentally helpful in persuading others, particularly inspectors and parents, of their professional competence. Further, their knowledge and understanding of research act as a marker of professional distinction when seeking promotion. Finally, one school leader reports how EEF recommendations were used to justify reducing the numbers of teaching assistants employed, despite this being an economic decision.

Benefits of and constraints on research use in Germany

Interestingly, for school leaders in Germany scientific knowledge is less important than the collaborative and collective contextual know-how relevant to the social setting:

If, for example, you ask someone else who is trusted to describe the situation, so that, so to speak, anonymously, 'I have a problem with the school management', 'I have a problem with a certain student group', 'I have a problem with a certain class', that can be very different then. And so you try to work with the know-how, what you already have at the school anyway, so of course that's not research-based, that's just more empirical ... in a very, very small frame ... But that, in any case, there must already have been investigations, because this seems to be relatively successful, so to speak, to find a colleague at all for solutions to certain problems.

(School Leader D)

Here, teachers trust their professional judgement without dismissing altogether the insights research might bring:

I know my class much better because I see them every day. I believe that science could make a contribution by adding to what is known, and perhaps it could go hand in hand. So research is often laughed at by the teachers, but actually ... it would be nice if somehow you could benefit from each other.

(School Leader E)

Although school leaders are cautious about what counts as research and what they believe is helpful, they feel recommendations from colleagues, advisors and consultants provide useful starting points. School Leader E, however, feels research such as Hattie and PISA results are difficult for individual schools to interpret. To overcome this, he uses an approach that he first encountered in initial teacher training called *Kollegiale Fallberatung*. This involves collegial case study discussions and exchanges about research, which are used to formulate recommendations and advice:

You could even get literature, it's not a problem ... in Hamburg there's the *Landesinstitut* ... but that would have to be done in your free time, and time is what you would need first.

(School Leader E)

Benefits of and constraints on research use in Scotland

School leaders in Scotland are largely negative about their experiences of the *EEF Toolkit*, one suggesting it is only ever used at a superficial level, if at all. School Leader H feels it is 'politically loaded' and potentially misleading, citing the example of teaching assistants who were described as a 'high-cost low impact' intervention:

We had our fairly new Director of Education saying things like 'I don't want to see that you're hiring classroom assistants in your PEF plans because ... that makes absolutely no difference whatsoever'. Now at ground level, we're saying 'that is nonsense, a good classroom assistant is worth their weight in gold'.

(School Leader H)

These school leaders do not trust research findings provided by intermediaries that contradict their personal experience. They also advocate approaching the *EEF Toolkit* with a degree of scepticism and considering how evidence relates to their context by asking, “is this actually relevant and right for my community?” (School Leader G).

For School Leader G, research is credible and useful when its implications for practice are clear to practitioners, adding that this also makes it easier for school leaders to demonstrate impact and encourage teacher ‘buy-in’: ‘I have tried to encourage the practical stuff. You catch more people if you help them to see the practical bit’ (School Leader G). School improvement programmes that are ‘academically rigorous’, as School Leader G puts it, are considered the most useful.

Further, research presented in an accessible form, not just to school leaders but to practitioners, is preferred over that from academic sources. Given practitioners’ workload, one school leader suggests that less academic resources written with practitioners in mind are easier for colleagues to digest. ‘There was this document that was a summary of the research evidence, it was tiny, very thin. That was great ... it was dead easy; it wasn’t hugely academic....’ (School Leader G). The status of the researcher also appears important:

Because you’ve got that expert, it was well received, he [William] is that knowledgeable expert... In my view, he has kept building on his understanding throughout his career. For me that gives him kudos, he knows what he’s talking about. He is really sure of himself, really confident. It’s confidence that has come from years of academic study.

(School Leader H)

But overall, while perceptions of academic rigour and whether the author is well-known and appears confident are taken as indicators of quality, these are less influential than clear connections to practice. Nevertheless, these school leaders are also aware that often research findings cannot simply be translated into practice, and that an understanding of how to adapt them to the context and community is important.

DISCUSSION: COMPARING SCHOOL LEADERS’ RESEARCH USE

In this paper we have explored how differences in national education policy contexts affect a small group of school leaders in each of three countries. Hence, the interpretations and suggestions that follow are offered cautiously.

It is perhaps inevitable that school leaders and teachers in England, faced with an imperative for school improvement in conditions of some jeopardy, are open to the help offered by interactional experts (Collins & Evans, 2007). Following years of government policy making, the research favoured fits a medical model, and practitioners attempt to marry clinical approaches to research use with functionalist approaches to school improvement. However, they lack both the time and research understanding to make good use of such advice. This supports the earlier conclusions of Allen et al. (2007). In any case, as Cain (2015) and others suggest, the translation of interactional expertise into practice is difficult, providing opportunities for research brokerage that predictably attract numerous state, civil, philanthropic and commercial intermediaries; too many, according to Siddiqui and Wardle (2020). While some select, synthesise, translate and publish research findings, others claim to use research to design resources that are marketed directly to schools or offer subscriptions to professional development and collaborative support. But school leaders and teachers are

so overwhelmed with suggestions and suspicious of big claims or assertions that contradict their professional experience that they depend instead on personal recommendation and networks. The upshot is that, somewhat ironically, school leaders and teachers find research most useful as an interactional tool to persuade others of their professional competence and distinction.

Although some of the features found in England apply to Scotland, albeit in a less risky context for school leaders and teachers, commercial involvement is limited and, although it is promoted by local authority and government agencies, schools are not compelled to use research. Consequently, teachers more often rely on their contributory professional expertise when deciding what is relevant and potentially useful. Scottish school leaders, like their English counterparts, value studies with clear links to practice but are also prepared to interpret and adapt research findings to fit school contingencies. This supports the findings of Cordingley (2000) and others. Practitioner research, which is included in the Professional Standards and widely used for school improvement, allows the interactional and contributory to inform each other. In contrast, only those with a specific interest or who are engaged in postgraduate study undertake practitioner research in England and Germany.

Finally, there is no impetus for research use by school leaders and teachers in Germany, where other national challenges such as migration and inertia in the school system dominate. A contributory model of practitioner expertise pertains as teachers make decisions about teaching and learning using a range of information to inform their professional judgement. Nevertheless, school leaders recognise that some insights from research might benefit practice, but their view is more liberal than medicalised accounts, where information and advice take a variety of forms and have implications that vary from school to school.

More broadly, our comparative analysis suggests that variations in approaches to research use between countries depend on the political context of schooling. Specifically, it matters whether the state is unified and relatively coherent or disaggregated through marketisation because the second allows many more interests to compete for the attention of school leaders and teachers, as schools themselves compete in the education marketplace. It matters whether schools are part of a stable but inflexible bureaucracy or a more uncertain market because the second affords a turn to managerialism and science in the search for predictability. It matters whether market accountabilities have low or high stakes for school leaders and teachers because the second encourages a strategic focus on maximising performance on valorised output measures, potentially putting practitioners' interests in conflict with those of some students. And finally, the extent to which there is public scrutiny of school leaders and teachers matters because this can erode public trust in professional judgement and provide an imperative for school leaders to justify decision making using disinterested research.

The school leaders' accounts provided here support Wilkins and Gobby's (2021) view that an instrumental-rational framing of research-use is inadequate for complex practice settings. They also fit Eyal's (2019) analysis that many of the difficulties with effective research use by school leaders and teachers are inherent and reflect a gap between the mechanical objectivity of research constructed to inform practice and the professional judgement of expert teachers. Marketisation affords the commodification of research, as both commercial learning science and brokerage intermediaries translate interactional expertise into resources, guidance and approaches that claim to improve school performance. Research is also used performatively by school leaders to justify their decision making when they are placed in positions of jeopardy and subject to public scrutiny. Such blatant interactional use is less common in lower-stake contexts, where practitioner inquiry and reflective practice allow a more productive relation between research findings and the craft knowledge of contributory expertise.

Limitations of the present study

This is a small-scale and exploratory study of three countries. Although we are confident that the documents analysed represent dominant policy positions in the selected territories, those in Germany are largely from one area and we have not explored regional variation. We conducted extended, in-depth interviews with a small number of school leaders. While this allowed us to faithfully construct detailed accounts of their understandings and experiences, they provide just a snapshot of the many possible perspectives of all school leaders, and we have not tested whether the views discussed here are representative of the wider group. Further, the views of school leaders may differ from those of their colleagues, be they fellow managers or teachers.

CONCLUSION

Our aim in this article has been to illuminate some of the challenges faced by school leaders in using research for school improvement as they move between policy requirements and the everyday complexities of professional practice. In this, our findings suggest school leaders' experiences of research use are shaped by political contexts and strongly framed by education governance processes. However, in contributing to the ongoing dialogue on evidence-informed practice in education, we acknowledge the tentative nature of these interpretations and so invite further exploration into the complex relationship between political contexts, education governance and research use.

Like Cairney and Geyer (2017) and Maxwell and her colleagues (2022), whose proposals we discussed earlier, we see a positive role for systems approaches to help make discussions between researchers, policy makers and practitioners more productive. Systems models start with preferred outcomes before identifying how these can best be realised. They understand that many different factors need to be aligned to achieve the desired results and their modelling of this is sophisticated. However, these idealised constructions of practice are based on assumptions about the capacities of individual actors that are undermined by sociocultural accounts of practice. Further, as idealisations they ignore the effects of power imbalances on actor behaviours. By talking to school leaders about their experiences of research use, we have begun to show how contributory experts struggle to combine broad and sometimes contrary suggestions from research with their professional understandings. However, we have found that practitioner research approaches can be helpful in bringing existing understandings and new knowledge together. We suggest systems accounts might incorporate similar collaborative and exploratory activities—where school leaders and teachers cooperate to adjust their practice as they make research findings their own—into their models but acknowledge that this would benefit from further study. By comparing experiences across national contexts, we have identified how everyday demands affect the capacity of school leaders to use research. Many of these demands, such as the impetus for research use by school leaders or research brokerage by intermediaries, originate at the national and regional level, begging the question of which national systems are more enabling. We suggest policies that increase professional insecurity and workload have a negative effect on research use but again recommend further exploration beyond this initial study.

Research should, we suggest, continue exploring the conditions that support effective research use and provide a broader analysis of policy enactment and its implications to include vanguard countries in practitioner research use like Australia and New Zealand. Nevertheless, our findings hint at what environments encouraging successful practitioner research use might look like. This picture aligns with accounts of the benefits of professional learning communities (OECD, 2015b, 2019b) and claims that collaborative school leadership

can improve teacher and student performance (Harris, 2013; Malloy & Leithwood, 2017). While collaboration allows school leaders and teachers to learn from and with each other, the interdependence provided by inclusive communities affords practitioners the security to do so. Although practitioner research set within such circumstances can take many forms (Chapman et al., 2016), we suggest these inquiries can be harnessed to explore, adapt, test and evaluate practical translations of existing research findings. Technology can assist, as communities of schools and wider networks support teacher research through information sharing, reading groups and discussions. Yet building collaborative school and online communities that support practitioner research use will be a challenge. How apt would it be if policy makers, practitioners and researchers worked together to achieve this.

AUTHOR CONTRIBUTIONS

Peter Kelly: Investigation; methodology; writing – review and editing; writing – original draft; formal analysis; conceptualization. **Anna Beck:** Investigation; methodology; writing – review and editing; formal analysis. **Susann Hofbauer:** Investigation; methodology; writing – review and editing; formal analysis.

CONFLICT OF INTEREST STATEMENT

There are no conflicts of interests to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

Ethical approval was given by the universities involved prior to the commencement of this study.

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Endnotes

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⁹ Originally <https://www.gov.uk/government/news/justine-greening-launches-new-institute-for-teaching>, Ambition School Leadership and the Institute for Teaching merged to become Ambition Institute, <https://www.ambition.org.uk/> (accessed 1 July 2023), whose courses are available for teachers at all stages of their career, including those wishing to become experts in their subject or wanting to learn the most effective ways to share best practice.

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- ¹⁴<https://www.forschungsmonitor-schule.de/> (accessed 1 July 2023).
- ¹⁵<https://digitallearninglab.de/> (accessed 1 July 2023).
- ¹⁶The health department of the State Institute for Teacher Education and School Development provides data collection, analyses and consulting for school leaders about mental stress and stress reduction: <https://li.hamburg.de/beratung/themen-aufgabengebiete/gefaehrungsbeurteilung/gefaehrungsbeurteilung-602230> (accessed 1 July 2023).
- ¹⁷<http://www.pathseducation.co.uk/> (accessed 1 July 2023).
- ¹⁸<https://www.gov.scot/groups/international-council-of-education-advisers/> (accessed 1 July 2023).
- ¹⁹<http://robertowencentre.academicblogs.co.uk/wp-content/uploads/sites/2/2017/06/Ebulletin-Issue-5-May-2017.pdf> (accessed 1 July 2023).
- ²⁰<http://www.tapestrypartnership.com> (accessed 1 March 2020).
- ²¹https://rccp.cornell.edu/downloads/TCI_7_SYSTEM%20BULLETIN.pdf (accessed 1 July 2023).
- ²²EBSCO is a division of the EBSCO Industries Inc., a large US company.
- ²³<https://education.gov.scot/improvement/> (accessed 1 July 2023).
- ²⁴<https://education.gov.scot/improvement/self-evaluation/interventions-for-equity/> (accessed 1 July 2023).
- ²⁵For example, ASPIRE Dundee, a community engagement programme, and a "Primary Literacy Coaching Programme" in Renfrewshire developed by an academic in a Scottish university.
- ²⁶<http://www.pedagoo.org/> (accessed 1 July 2023).
- ²⁷<https://www.pioneerap.co.uk/> (accessed 1 July 2023).
- ²⁸<https://arkonline.org/> (accessed 1 July 2023).
- ²⁹<https://www.arkcurriculumplus.org.uk/mastery> (accessed 1 July 2023).
- ³⁰<https://www.challengepartners.org/> (accessed 1 July 2023).
- ³¹<https://www.challengepartners.org/challenge-gap> (accessed 1 July 2023).
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APPENDIX 1

SCHOOL LEADER INTERVIEW QUESTIONS

What information do you use in informing your decisions?

Where does this come from?

Is there any advice available on this?

How important is education research to you?

Do you believe education researchers?

Are there some you believe more than others?

What gives you confidence in their work?

What research have you heard or read about recently?

What kinds of research do you find interesting and why?

Can you give an example?

How do you find out about research developments?

How easy is this?

What helps with this and what are the barriers?

Where are the places you most go to find about research?

How easy is this?

How clearly is the research presented?

Do you use research findings to inform your practice?

How? Can you provide some examples?

Do you or your colleagues do any research yourselves?

How does this help to inform practice?

Can you give an example?

Do you know of any national or local government policies based on research?

How did they affect you?

Can you give an example?

Have you initiated any change or developments based on research?

How well did these work?

Do you talk to teachers in your school about research?

What kinds of things do you find interesting?

How do they feel about research?

What about other principals?

Where do you agree and where do you differ?